

Final Project Report

Predicting Patient Satisfaction from Hospital Reviews Using Traditional Machine Learning

IE 7275: Data Mining in Engineering

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GitHub repository: <https://github.com/Wiktoria0110/IE7275-Data-Mining-Project/tree/main>

1. Problem Setting

Access to quality healthcare is a growing challenge in Canada and worldwide. Patients who receive medical care frequently share their experiences through online reviews, providing hospitals and healthcare administrators with a valuable but largely underutilized source of feedback. Healthcare providers receive hundreds or thousands of these reviews but lack the time and automated tools to analyze them systematically at scale (Greaves et al., 2013). The inability to process patient feedback efficiently means that recurring issues - such as long wait times, poor communication between staff and patients, or inadequate treatment - often go unaddressed. An automated system capable of classifying patient sentiment would enable healthcare providers to identify problem areas quickly, prioritize improvements, and ultimately deliver better patient care. Data mining techniques offer a practical and scalable approach to this challenge, enabling pattern recognition across large volumes of unstructured text data (Manning et al., 2008).

2. Problem Definition

This project addresses the following core problem: given a hospital patient review written in natural language, can a machine learning model automatically and accurately predict whether the patient experience was positive or negative? More specifically, the project aims to answer the following research questions: RQ1: Can traditional machine learning classification models accurately predict patient satisfaction (positive or negative) from hospital review text? RQ2: Which classification algorithm - among Logistic Regression, Naive Bayes, Decision Tree, Support Vector Machine, and K-Nearest Neighbors - performs best on this task? RQ3: What textual features are most predictive of positive or negative patient experiences? The outcome is a practical tool that healthcare providers can use to automatically monitor patient feedback at scale, identify recurring issues, and make data-driven decisions to improve quality of care.

3. Data Sources

The primary dataset used in this project is the Hospital Patient Reviews Dataset, publicly available on Kaggle (Hospital Reviews Dataset, 2023). The dataset contains real patient reviews of hospitals along with sentiment labels and star ratings. It is freely available, contains no personally identifiable information, and raises no ethical concerns. Dataset URL: <https://www.kaggle.com/datasets/junaid6731/hospital-reviews-dataset>

4. Data Description

The Hospital Patient Reviews Dataset contains 996 patient reviews across 3 columns. Table below provides a detailed description of the dataset.

Attribute	Detail
Dataset name	Hospital Patient Reviews Dataset
Number of rows	Kaggle
Number of columns	966 patient
Column1: Feedback	3 columns
Column 2: Sentiment label	Free-text patient review
Column 3: Ratings	Binary label: 0 – negative, 1 – positive (target variable)
Missing values	0
Label distribution	728 (positive (73.1%), 268 negative (26.9%) - imbalanced
Train/Test split	796 training, 200 test, 80/20

An important characteristic of this dataset is its class imbalance - 73.1% of reviews are positive and only 26.9% are negative. This reflects the real-world distribution of hospital reviews, where patients who had strongly negative experiences are less likely to leave

written feedback. This imbalance has significant implications for model evaluation, as discussed in Section 8.

5. Data Exploration

Exploratory data analysis was conducted to understand the structure, distribution, and characteristics of the dataset before applying machine learning models.

5.1 Missing Value Analysis

A missing value check was performed across all three columns. No missing values were found in any column (total missing = 0). This means no imputation or data cleaning for missing values was required, and all 996 reviews could be used for modelling.

5.2 Label Distribution

The dataset contains 728 positive reviews (73.1%) and 268 negative reviews (26.9%). This imbalanced distribution is typical of real-world review datasets, where patients who had positive experiences are more likely to leave feedback. The imbalance was addressed through the use of F1-score as the primary evaluation metric and stratified train/test splitting.

5.3 Review Length Analysis

Review length was analysed to understand the nature of patient feedback and guide feature engineering decisions.

Statistic	Value
Mean length	20.48 words
Median length	19 words
Standard deviation	10.47 words
Minimum length	1 word
Maximum length	149 words

Hospital reviews are notably short - with a mean of only 20 words per review. This is significantly shorter than typical product or restaurant reviews. This characteristic directly influenced the TF-IDF feature selection strategy - a smaller max_features value of 500 was chosen, appropriate for the limited vocabulary size of the dataset.

6. Data Mining Tasks

The following data mining tasks were performed in this project:

6.1 Data Preprocessing

Text preprocessing was applied to clean the raw review text before feature extraction. The preprocessing pipeline included: (1) converting all text to lowercase to ensure consistency, (2) removing punctuation and special characters using regular expressions, (3) removing stop words - common English words such as 'the', 'a', 'is', and 'was' that carry no sentiment information - using the NLTK library, and (4) tokenization to split text into individual words.

6.2 Feature Extraction - TF-IDF Vectorization

TF-IDF (Term Frequency-Inverse Document Frequency) vectorization was applied to convert cleaned text into numerical feature vectors (Salton & Buckley, 1988). TF-IDF assigns each word a weight based on how frequently it appears in a review relative to how common it is across all reviews. Words that appear frequently in one review but rarely across all reviews receive higher weights. A maximum of 500 features was selected, appropriate for the small vocabulary size of the dataset (mean review length: 20 words). The vectorizer was fitted on training data only and applied to test data to prevent data leakage.

6.3 Dimension Reduction - Chi-Squared Feature Selection

Chi-squared feature selection was applied to reduce the 500 TF-IDF features to the 200 most statistically significant features (Manning et al., 2008). The chi-squared test measures the statistical association between each word feature and the target class

label. Words that appear more frequently in one class (positive or negative) receive higher chi-squared scores and are retained. Words that appear equally in both classes are removed as they provide no discriminative information. This dimension reduction step reduces noise, improves computational efficiency, and helps prevent overfitting.

6.4 Binary Classification

The primary data mining task is binary classification - predicting whether a review is positive (1) or negative (0). Five classification algorithms were trained and compared, as described in Section 7.

6.5 Model Validation - 5-Fold Cross Validation

5-fold cross validation was applied to all models to provide reliable performance estimates. The training set (796 reviews) was divided into 5 equal folds. Each model was trained on 4 folds and evaluated on the remaining fold, repeating this process 5 times. The mean F1 score across all 5 folds was reported. This approach is particularly important given the small dataset size, as a single train/test split may produce unreliable estimates.

7. Data Mining Models and Methods

Five traditional machine learning classification algorithms were implemented and compared. All models used the same 200 chi-squared selected TF-IDF features as input.

7.1 Logistic Regression

Logistic Regression is a linear classification model that learns a weight for each input feature through supervised training. For text classification, it learns which words are most strongly associated with positive or negative sentiment. The model was trained with `max_iter=1000` and `random_state=42` for reproducibility. In addition to classification, the learned coefficients provide interpretability - the top positive and negative words can be extracted and visualized.

7.2 Naive Bayes

MultinomialNB was used, which is specifically designed for word count features such as TF-IDF. Naive Bayes calculates the probability that a review belongs to each class based on the words it contains, assuming word independence. Despite this simplistic assumption, Naïve Bayes is known to perform well on text classification tasks and is computationally very efficient.

7.3 Decision Tree

Decision Tree classifier builds a hierarchical series of yes/no questions to classify reviews. At each node, the algorithm selects the feature that best separates the classes. While interpretable, Decision Trees are prone to overfitting - especially on small datasets - as they tend to memorize training examples rather than learning general patterns.

7.4 Support Vector Machine (SVM)

SVM with a linear kernel was used (Joachims, 1998). SVM finds the optimal boundary between positive and negative reviews by maximizing the margin between the two classes. SVM is particularly well-suited for high-dimensional text data and small datasets, making it a strong candidate for this project. The linear kernel was chosen as TF-IDF features are already high-dimensional and linearly separable.

7.5 K-Nearest Neighbors (KNN)

KNN with $k=5$ classifies each review by finding the 5 most similar reviews in the training set and taking a majority vote. Similarity is measured using distance between TF-IDF vectors. KNN is sensitive to class imbalance as the majority class tends to dominate the neighborhood.

8. Performance Evaluation

Multiple evaluation metrics were used to comprehensively assess model performance.

8.1 Evaluation Metrics

Accuracy: The proportion of correctly classified reviews out of all reviews. While simple to interpret, accuracy is misleading for imbalanced datasets - a model that always predicts positive would achieve 73% accuracy without learning anything useful. **F1-Score (Primary Metric):** The harmonic mean of precision and recall. F1 was designated as the primary metric because it accounts for class imbalance by balancing both precision and recall. A model that only predicts the majority class would receive a low F1 score, making it a more reliable metric for this dataset.

Precision: Of all reviews predicted as a given class, what proportion were actually that class? High precision means few false positives.

Recall: Of all actual reviews belonging to a given class, what proportion were correctly identified? High recall means few false negatives.

ROC-AUC: The Area Under the Receiver Operating Characteristic curve measures the model's ability to discriminate between classes across all possible thresholds. A value of 1.0 indicates perfect discrimination; 0.5 indicates random guessing.

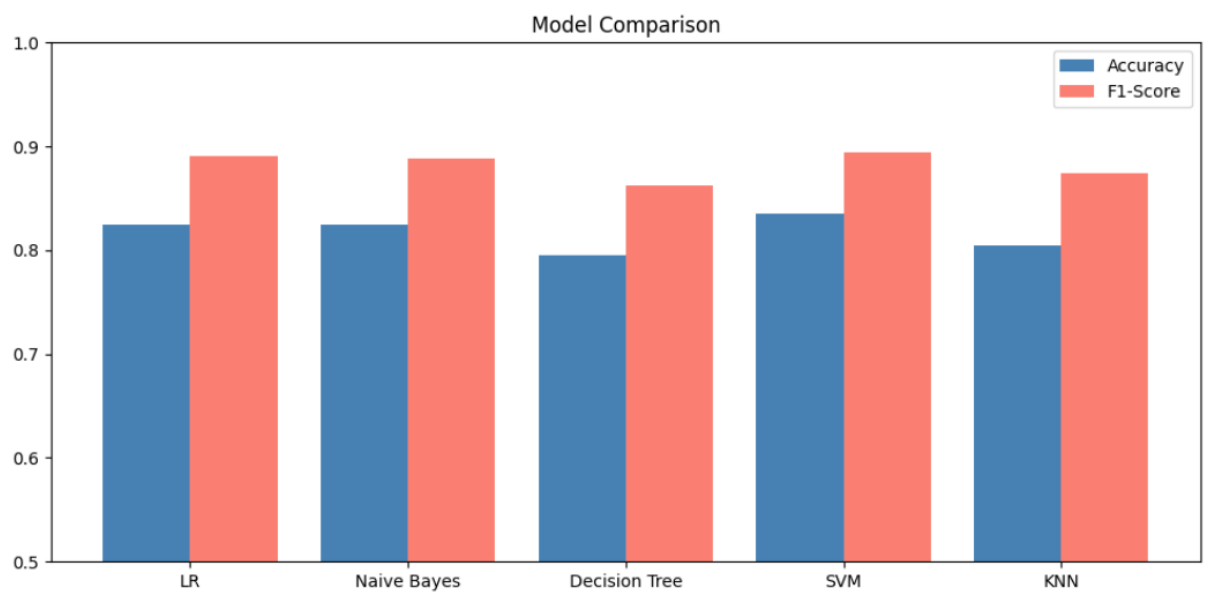
5-Fold Cross Validation Mean F1: The average F1 score across 5 cross-validation folds provides a more reliable performance estimate than a single train/test split, especially important given the small dataset size.

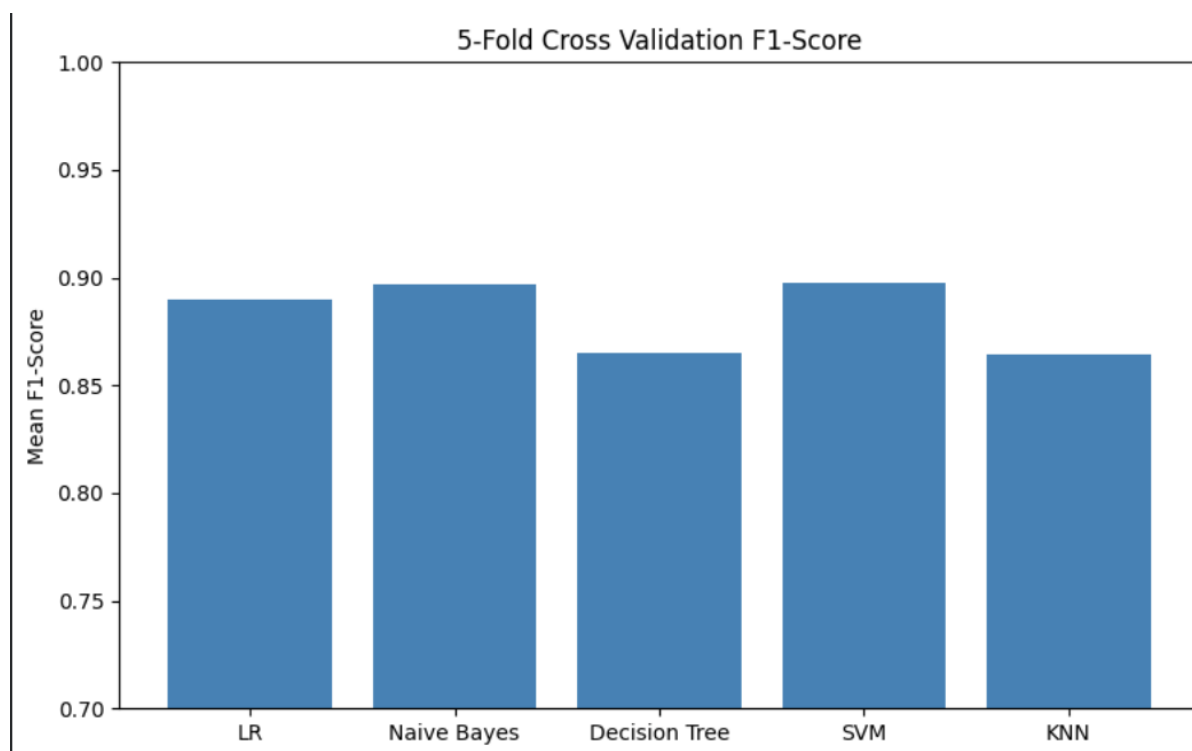
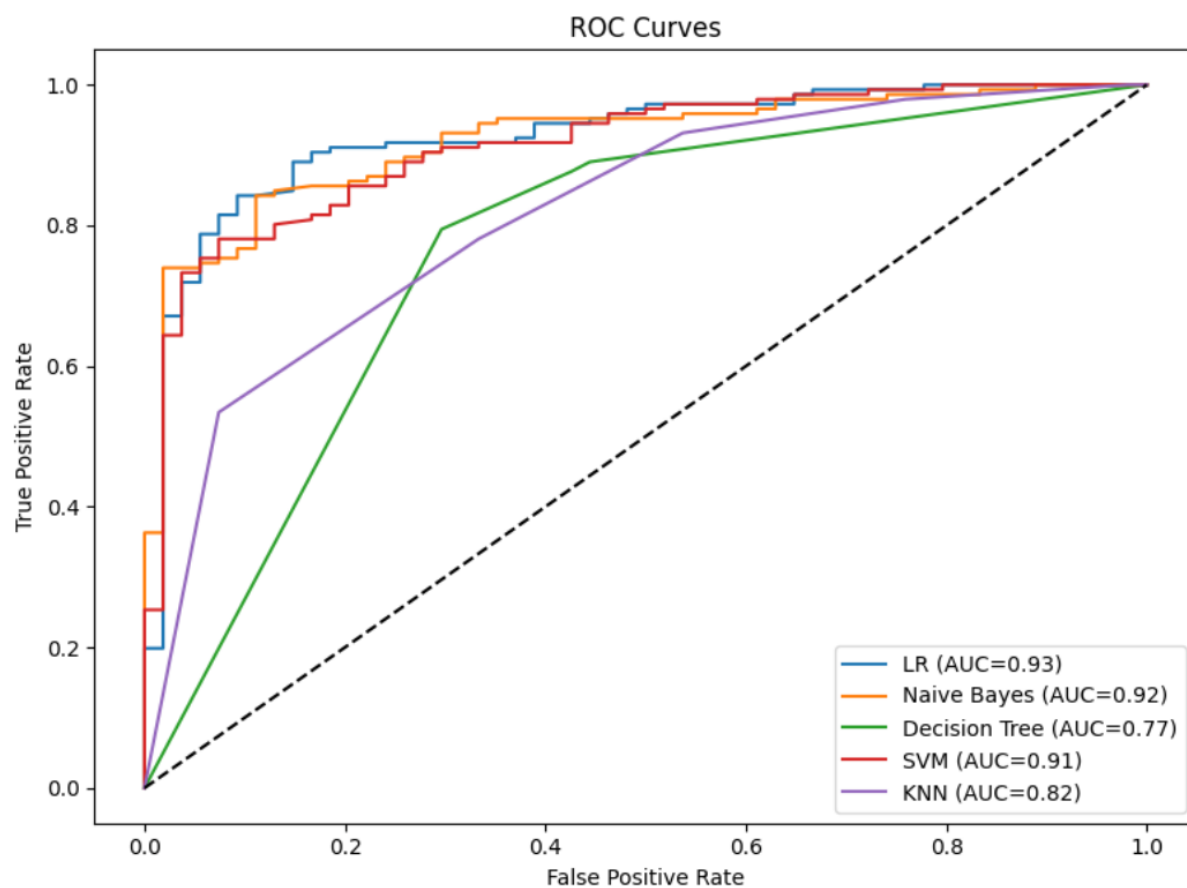
9. Project Results

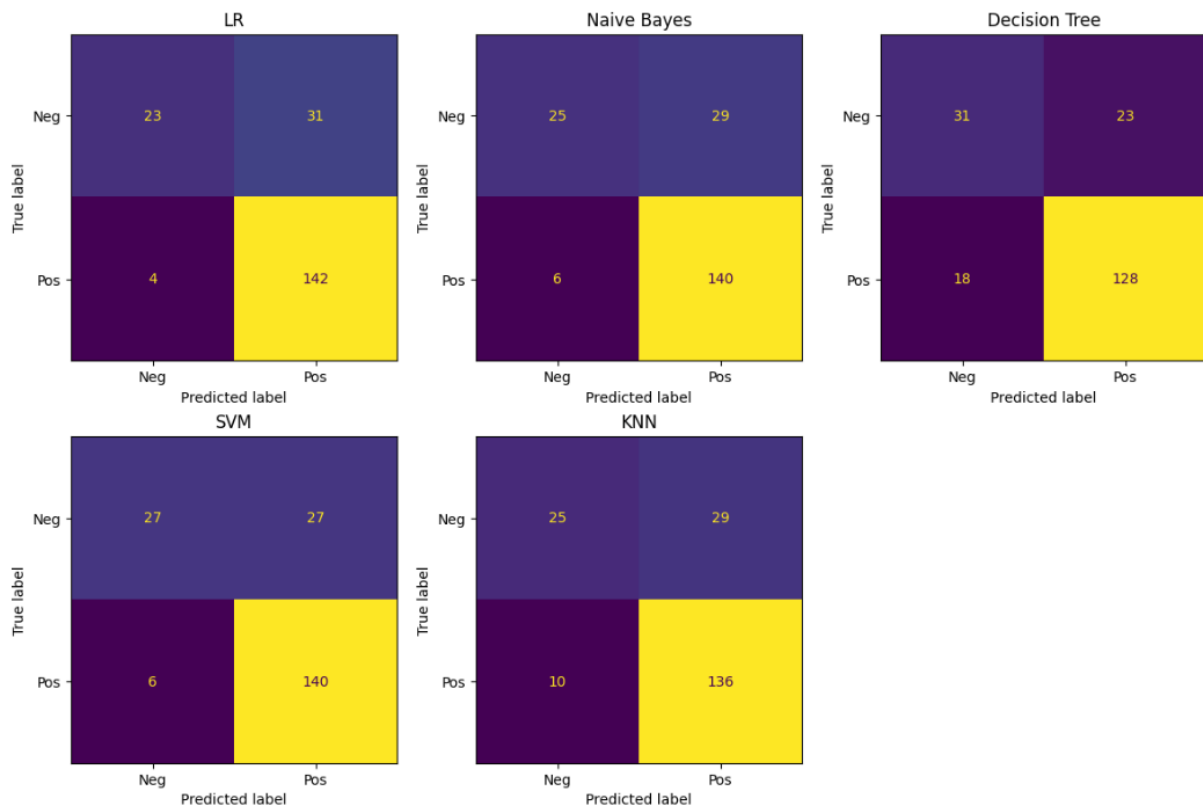
9.1 Model Performance on Test Set

Table below presents the performance of all five models on the held-out test set (200 reviews).

Model	Accuracy	F1-Score	CV Mean F1	ROC-AUC
Logistic Regression	82.50%	89.03%	88.98%	0.93
Naïve Bayes	82.50%	88.89%	89.71%	0.92
Decision Tree	79.50%	86.20%	86.53%	0.77
SVM	83.50%	89.46%	89.74%	0.91
KNN	80.50%	87.46%	86.40%	0.82







9.2 Key Findings

SVM achieved the best overall performance with 83.50% accuracy and 89.46% F1-score on the test set, and the highest cross-validation mean F1 of 89.74%. This confirms the hypothesis stated in the project proposal. SVM's effectiveness for text classification tasks is well-established in the literature (Joachims, 1998), particularly for high-dimensional feature spaces like TF-IDF.

All models struggle with the negative class - recall for class 0 (negative) ranges from 0.43 to 0.57 across all models. This is a direct consequence of class imbalance: with only 27% negative reviews in the training set, models tend to predict positive when uncertain. This is reflected in the confusion matrices, where all models correctly identify the majority of positive reviews but miss a significant proportion of negative reviews.

Decision Tree performs worst with 79.50% accuracy and 86.20% F1-score. This is expected - Decision Trees are prone to overfitting on small datasets by memorizing training examples rather than learning generalizable patterns.

Logistic Regression achieves the highest ROC-AUC (0.93) despite SVM having the highest F1-score. This indicates that LR is better at ranking predictions across all thresholds, while SVM is better at the specific classification threshold of 0.5.

Cross-validation confirms SVM as the most reliable model with mean F1 of 0.8974 and standard deviation of 0.0185 - showing consistent performance across different data splits.

9.3 Per-Class Analysis

Table below shows the per-class precision and recall for the best performing model (SVM).

Class	Precision	Recall	F1-Score	Support
0 (negative)	0.82	0.50	0.62	54
1 (positive)	0.84	0.96	0.89	146

The SVM model achieves high recall for positive reviews (0.96) but lower recall for negative reviews (0.50). This asymmetry is due to class imbalance - the model correctly identifies 96% of positive reviews but only 50% of negative reviews. Precision for both classes is similar (0.82-0.84), indicating the model makes few false positive errors when it does predict a class.

10. Impact of Project Outcomes

The data mining pipeline developed in this project has significant practical value for healthcare providers and hospital administrators.

Automated feedback monitoring: The SVM model can automatically classify thousands of patient reviews in seconds - a task that would take human administrators hours or days to complete manually. This enables real-time monitoring of patient satisfaction trends.

Data-driven quality improvement: By automatically identifying negative reviews, hospital management can quickly pinpoint recurring issues - such as long wait times, poor communication, or inadequate treatment - and take targeted corrective action.

Scalability: The pipeline can be applied to any hospital or clinic review dataset with minimal modification, making it a generalizable tool for the healthcare industry.

Interpretability: The Logistic Regression model provides additional value through its learned word coefficients, which reveal which specific words and phrases most strongly indicate patient satisfaction or dissatisfaction - providing actionable linguistic insights for hospital communication teams.

Limitation and future work: The main limitation is the class imbalance (73/27) which reduces the model's ability to detect negative reviews. Future work should explore oversampling techniques such as SMOTE, cost-sensitive learning, and ensemble methods to improve negative class detection. Additionally, testing on larger and more diverse hospital review datasets would strengthen the generalizability of the findings.

References

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